

Lab 6: MOSFETs

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This lab investigates the behavior MOSFETs. A MOSFET is a semiconductor primarily used for amplifying signals where a small base current controls a collector current to amplify signal strength. A MOSFET consists of a source, drain, and gate terminal. Between the source and drain lies a doped semiconductor region. Above this channel is a thin insulating oxide, typically SiO_2 , and on top a conductive gate material that is usually silicon or a metal. Since the gate is insulated, virtually no current flows into it. When a gate-to-source voltage is applied an electric field forms an inversion layer at the semiconductor-oxide interface. This creates a conductive channel that allows carriers, electrons in an n-channel MOSFET or holes in a p-channel MOSFET, to flow. Moreover, the ability to precisely control channel conductivity through an electric field, without requiring gate current, enables MOSFETs to achieve high input impedance, low power consumption, and scalability essential for modern integrated circuits. This makes MOSFETs ideal for switching and amplification, especially in digital logic circuitry where minimal gate current is essential.

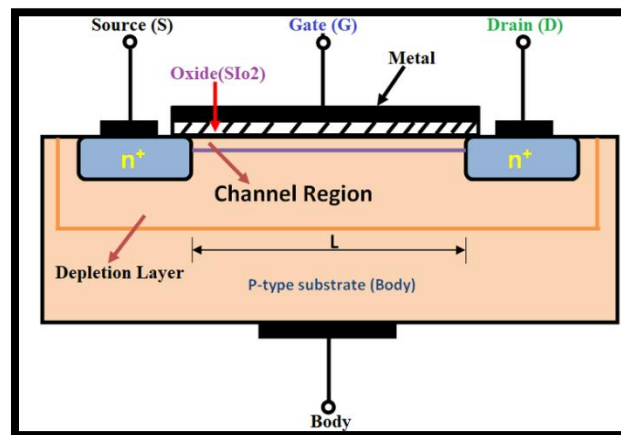
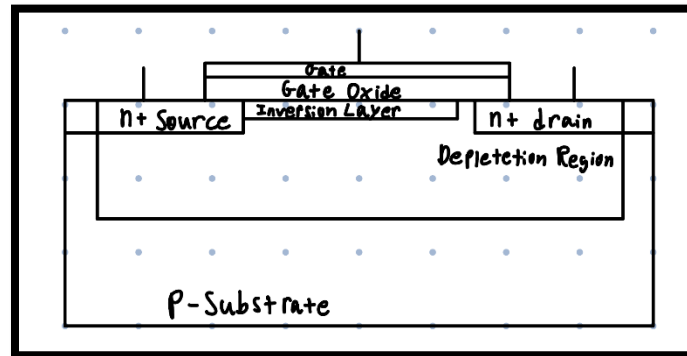


Figure 1: MOSFET Structure

Fundamentals:

Sketch:

*Figure 2: MOSFET Inve*

A depletion-mode transistor is a device that conducts at zero gate bias because a conductive channel already exists. Applying a negative gate voltage will deplete carriers from this channel and reduce the drain current. In other words, the channel is on and the gate voltage must be used to turn the device off. This contrasts with other MOSFET devices, which require a positive V_{GS} to create the channel.

Mathematical definitions of drain conductance and transconductance of a MOSFET:

$$g_d = \frac{\partial I_D}{\partial V_{DS}} \Big|_{V_{GS} = \text{constant}}$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}} \Big|_{V_{DS} = \text{constant}}$$

Experiment 1: Demonstrate the Regimes of Operation of a MOSFET

The goal of experiment 1 is to measure the different operating modes of a MOSFET. A 5LN01SP N-type transistor is used. A Keysight U2722A SMU and its AMM software is utilized to record measurements. The setup is simple. The MOSFET is connected to a breadboard and wired to the SMU. Sweeps are set in the software corresponding to each operating mode the MOSFET can be in. The IV curves are plotted and displayed below:

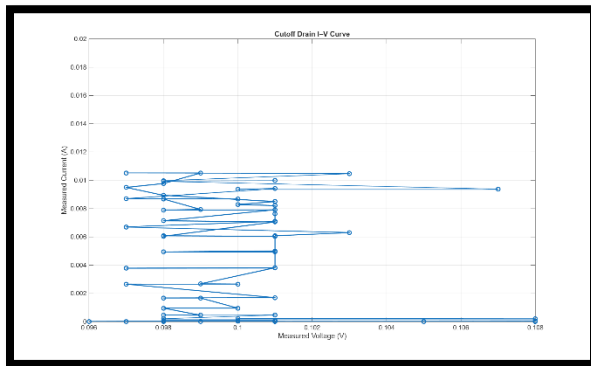


Figure 3: Cutoff Drain IV Curve

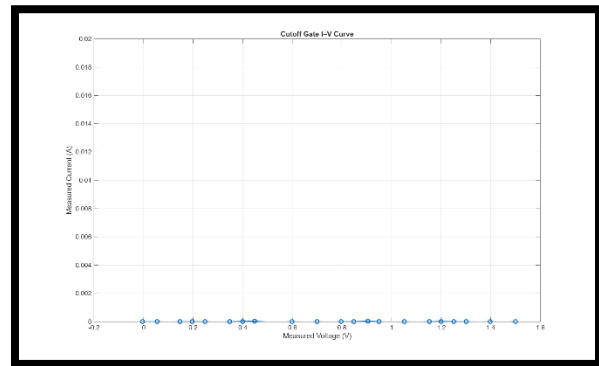


Figure 4: Cutoff Gate IV Curve

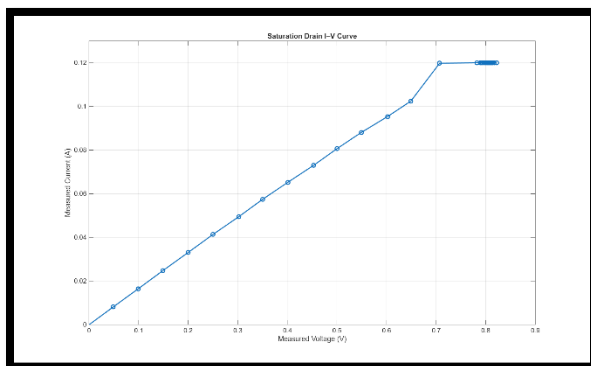


Figure 5: Saturation Drain IV Curve

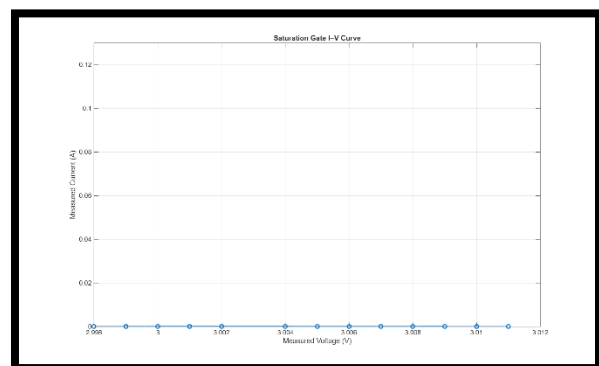


Figure 6: Saturation Gate IV Curve

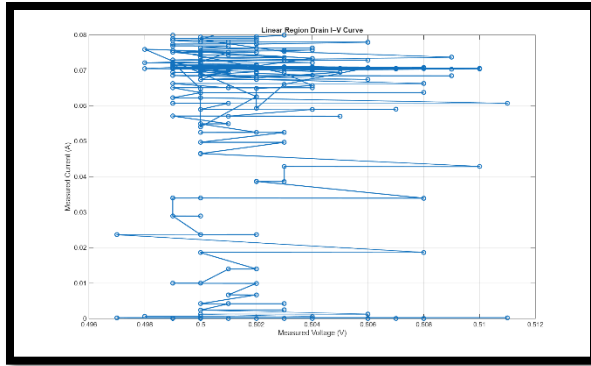


Figure 7: Linear Drain IV Curve

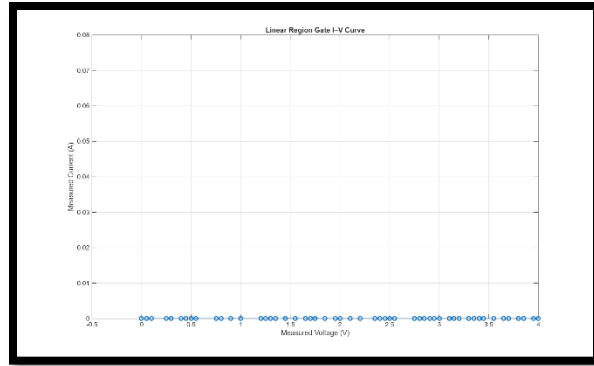


Figure 8: Linear Gate IV Curve

Concept Question(s):

According to the recorded data, each region is relatively ideal except for the linear region. The voltage did not sweep causing multiple noisy measurements at relatively same voltage. The cut-off region performs as expected with roughly zero measurement besides some noise on the drain's IV curve. For the linear region, I should have seen a clean and slightly curved IV curve as V_{DS} keeps increasing instead of a bunch of disorganized data points.

Experiment 2: Demonstrating the Forward Active Mode of a MOSFET

The goal of experiment 2 is to duplicate a series of curves for a MOSFET in forward active mode to understand the behavior of transitioning from ohmic to saturation regimes. I_D is measured versus V_{DS} curves for different values of V_{GS} .

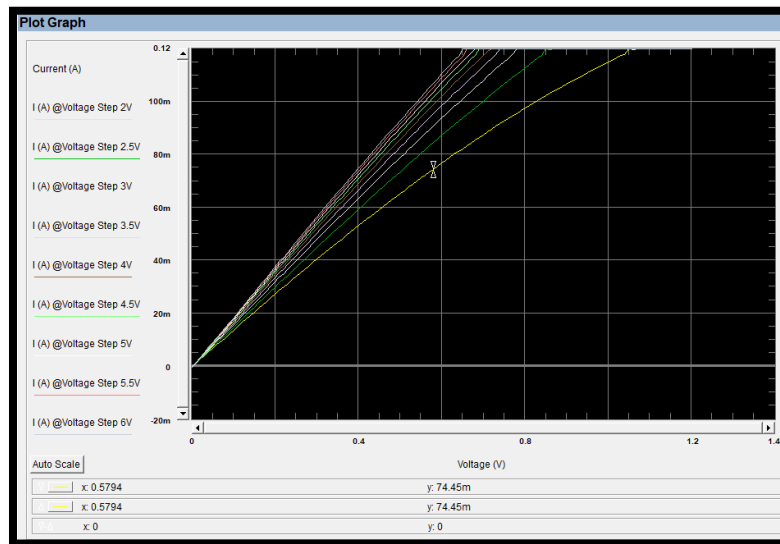


Figure 9: Series of Curves

A separate sweep with the SMU was not run after generating the curves graph. So the results can't be fitted to the square law directly. However, due to the perfect series of curves, it should fit extremely well with the square-law theory. In forward-active mode, the drain current should be described by the square-law equation:

$$I_D = \frac{1}{2} k (V_{GS} - V_{th})^2$$

It estimates that the saturation current should quadratically increase with the gate voltage. It also shows that each successive curve should be higher than the last. So even though the series of curves never reached a high level of saturation, they still illustrate the correct slope and separation behavior.

So although I do not have the full data and parameters for Method B of this experiment, the initial curve test of this device suggests that it should behave consistently with the square-law theory.

Concept Question(s):

When compared to the MOSFET's datasheet, it behaves exactly as expected. The curves line up well with the charts given.

Experiment 3: Switching a Device On and Off with a MOSFET

The goal of experiment 3 is to create a simple switching circuit that can turn an LED on and off. Various measurements are recorded such as the power consumed by the power line. The setup utilizes a waveform generator and an oscilloscope as well as creating the circuit below with a ZVN2110a MOSFET.

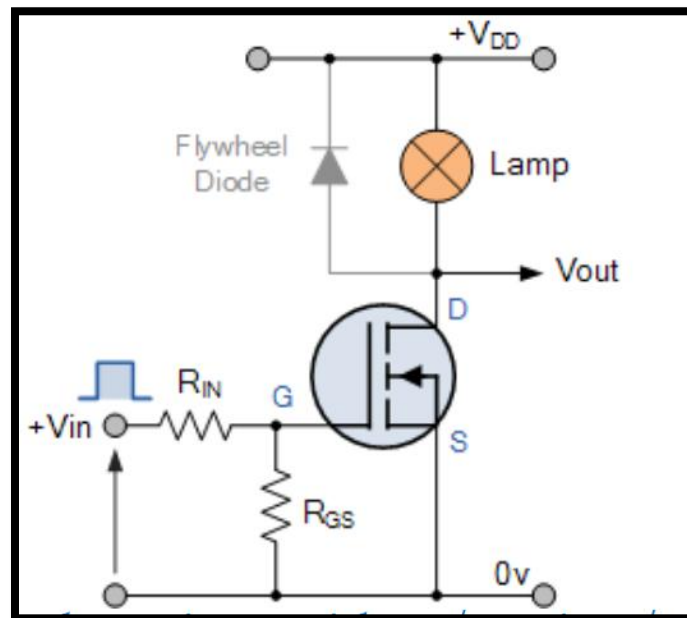


Figure 10: Switching Circuit

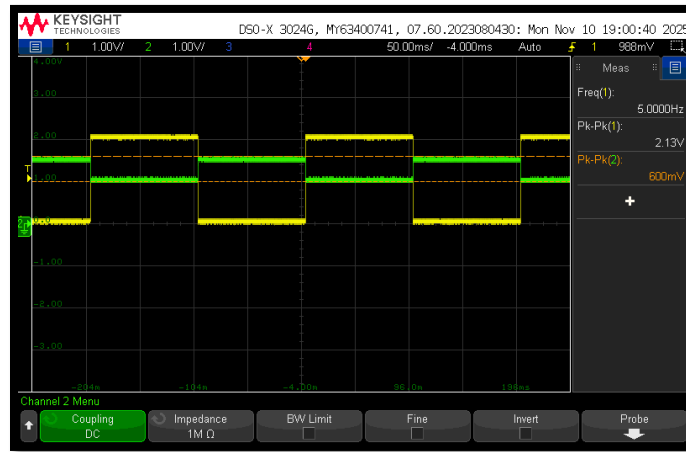


Figure 11: Oscilloscope Capture

A $10\ \Omega$ sense resistor is used. The current is calculated using the resistor value and green waveform (Square wave, Frequency: 5 Hz, Amplitude: 1 V Pk-to-Pk, Offset: 0.5 V, Duty Cycle 50%).

Peak LED Current:

$$I_{Peak} = \frac{V_{High}}{R} \approx \frac{0.6\ V}{10\ \Omega} = 0.06\ A = 60\ mA$$

Average LED Current (50% duty cycle):

$$I_{Average} = D \times I_{Peak} = 0.5 \times 60\ mA = 30\ mA$$

RMS Current:

$$I_{RMS} = I_{Peak} \sqrt{D} = 60\ mA \sqrt{0.5} \approx 42\ mA$$

Experiment 4: Amplifying a signal with an N-channel MOSFET

The goal of experiment 4 is to amplify a waveform signal with an N-Channel MOSFET. The circuit is also used to power an audio speaker. Oscilloscope shots are captured to determine how well the output matches the input sine wave. The setup involves constructing the circuit below:

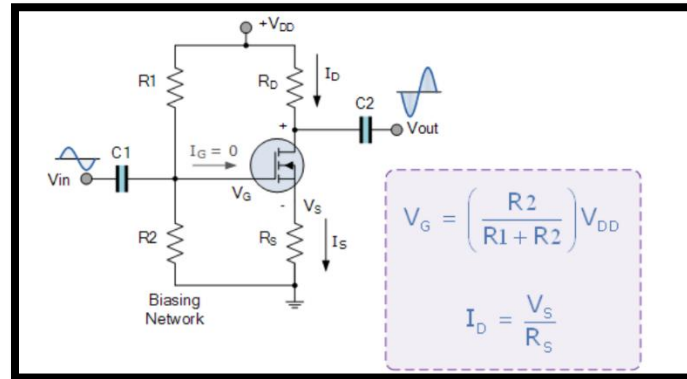


Figure 12: Common Source Amplifier



Figure 13: Method B Waveform

Parameters:

- $R_1 = 470 \text{ k}\Omega$, $V_{DD} = 10 \text{ V}$, $R_2 = 220 \text{ k}\Omega$, $R_S = 1 \text{ k}\Omega$, $R_D = 2.2 \text{ k}\Omega$, $C_1 = C_2 = 1 \text{ }\mu\text{F}$
- Channel 1 Sine Wave: Frequency: 1 kHz, Amplitude: 1 V Pk-to-Pk, Offset: 0 V
- Channel 2: Drain output. Frequency 1 kHz, Measured Pk-to-Pk: 4.8 V

Voltage Gain:

$$A_v = \frac{V_{out,pk-pk}}{V_{in,pk-pk}} = \frac{4.8\text{ V}}{2.2\text{ V}} = 2.2\text{ V}$$

The amplifier has a voltage gain of $\sim -2.2\text{ V}$ in magnitude (negative due to the inverted output). When observing the image, the yellow sine wave is at its maximum while the green one is at its minimum.

Extra Credit:

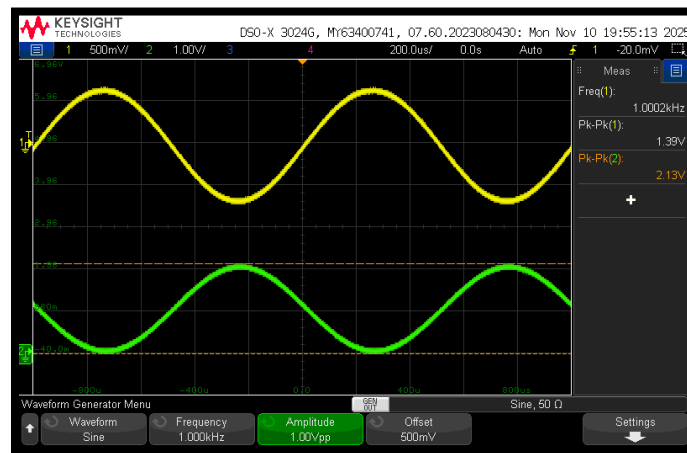


Figure 14: Extra Credit Waveform

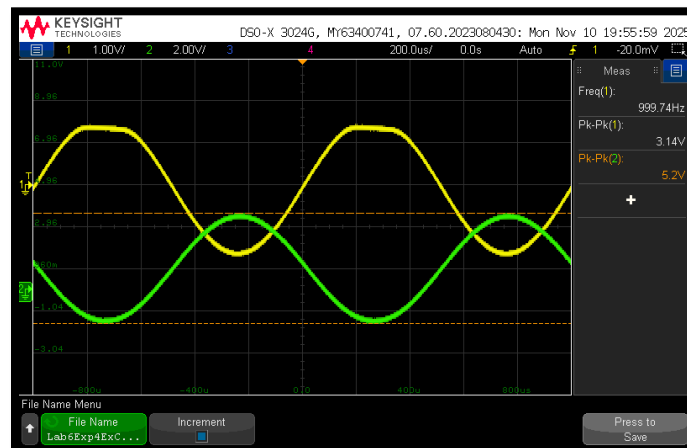


Figure 15: Extra Credit Waveform 2.5 V Pk-to-Pk

Parameters (Figure 14):

- Channel 1 Sine Wave: Frequency: 1 kHz, Amplitude: 1.39 V Pk-to-Pk, Offset: 0 V
- Channel 2: Speaker output. Frequency 1 kHz, Measured Pk-to-Pk: 2.13 V

Voltage Gain:

$$A_v = \frac{V_{out,pk-pk}}{V_{in,pk-pk}} = \frac{2.13 \text{ V}}{1.39 \text{ V}} = 1.53 \text{ V}$$

Parameters (Figure 15):

- Channel 1 Sine Wave: Frequency: 1 kHz, Amplitude: 3.14 V Pk-to-Pk, Offset: 0 V
- Channel 2: Speaker output. Frequency 1 kHz, Measured Pk-to-Pk: 5.2 V

Voltage Gain:

$$A_v = \frac{V_{out,pk-pk}}{V_{in,pk-pk}} = \frac{5.2 \text{ V}}{3.14 \text{ V}} = 1.66 \text{ V}$$

The MOSFET amplifier circuit was used to drive a small speaker using a 1 kHz sine wave. The oscilloscope plots show that the input (yellow wave) and the speaker output (green wave) are both nearly 1 kHz each. This was run for two amplitude inputs to try and get a larger gain. It proves that the circuit does not significantly affect the output frequency out of the speaker. For a 3.14 V_{pk-to-pk}, the gain can be about ~1.66 V while maintaining a clear 1 kHz tone. This was confirmed using the Sonic Tools iOS app to read the frequency as well.

Concept Question(s):

In this common-source amplifier, the gain is limited by the MOSFET's transconductance g_m and the drain resistor R_D . Increasing the gain would need a large R_D and g_m . This can be done if the MOSFET is biased at a higher drain current. If this were tried though, the higher R_D would reduce the drain voltage and the MOSFET would most likely exit its saturation state. Since our voltage supply is only 10 V, the output will clip if the gain is exceedingly high regardless. With these chosen resistor values, the measured gain is exactly what is expected. If the gain is pushed further than the 1.66 V found in the speaker extra credit, the output wave would most likely be distorted. So a moderate gain can be achieved, but to go further, a different setup would be required with a different MOSFET and biasing parameters.

The two capacitors used act as high-pass filters for the input and output. With the chosen component values, this input and output filter should cut off at about 70 Hz. This means that lower frequency signals will have a reduced gain compared to higher frequencies. But at very high frequencies, the MOSFET will actually end up reducing the gain and start to introduce a distorted shift.

Conclusion:

Across these four experiments I gained a much better understanding of how MOSFETs operate in both switching and amplification roles. By characterizing the device in the cutoff, linear, and saturation regions, I learned how the gate-to-source voltage controls the channel and how the drain current behaves when biased differently. The switching experiment demonstrated how in MOSFETs: the gate draws essentially no steady-state current, allowing the device to control relatively large drain currents with minimal input. The fourth experiment, building the common-source amplifier, load resistors and coupling capacitors alter the circuit's gain and frequency response. When compared to the previous BJT lab, it is clear the similarities between the two devices. Overall, this lab strengthened my ability to analyze MOSFET behavior, troubleshoot switching/amplifying circuits, use laboratory instruments, and connect theoretical models to real-world behavior.

GenAI Usage:

AI was used to help plot the experiment 1 data. It had to be altered to correct the scaling that it kept outputting, however. It also helped to clearly explain the differences between the different types of MOSFETs.

Citations:

MOSFET Structure Image:

Venn, Matt. "MOSFET." *Zero to ASIC Course*, 12 Nov. 2020,

www.zerotoasiccourse.com/terminology/mosfet/